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Task 1:

TT:

A1	A0	B1	B0	L (A<B)	E (A=B)	G (A>B)
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1

A1	A0	B1	B0	L (A<B)	E (A=B)	G (A>B)
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

Expression:

Simplified expression

Equal - Both equal

$$E = (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$

Greater $A_1 > B_1$ or $A_1 = B_1$ and $A_0 > B_0$

$$G = A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0$$

Less than $A_1 < B_1$ or $(A_1 = B_1$ and $A_0 < B_0)$

$$L = \bar{A}_1 B_1 + (A_1 \odot B_1) \bar{A}_0 B_0$$

Circuit Diagram:

